



FUZZY

Inverted Pendulum Fuzzy Logic Controller Simulation

This report presents the design, implementation, and evaluation of a fuzzy logic controller for an inverted pendulum system. The inverted pendulum is a classical benchmark problem in artificial intelligence and control engineering due to its nonlinear dynamics and inherent instability.

1. Introduction

The inverted pendulum system is made of a strict pole that is fixed to a moving cart. The goal is to hold the pole in an upright position using a suitable horizontal force to the cart. In the absence of active control, the system is in a state of instability and the pole will drop very fast due to gravity.

This issue has been common in control engineering education since it allows the investigation of the main issues like nonlinearity, instability, and the necessity of real-time feedback control.

2. Project Objectives

The objectives of this assignment are to:

- Implement a dynamic simulation of an inverted pendulum.
- Design and apply a fuzzy logic controller.
- Demonstrate stabilization for small to moderate disturbances.
- Generate and analyze a fuzzy control surface.
- Discuss system limitations.

3. System Model

There are four main variables that describe the system state; cart position, cart velocity, pole angle, and pole angular velocity. The controller provides a horizontal force which acts on the cart.

4. Dynamics of the System

The inverted pendulum mathematical model is based on Newtonian mechanics. Horizontal acceleration and angular acceleration are calculated using coupled nonlinear equations which incorporate gravitational, inertial and control forces.

A fixed simulation time step is used to integrate the equations numerically and update the system state during execution.

5. Fuzzy Logic Controller

The fuzzy controller takes two inputs, where pole angle and angular velocity are the inputs. These variables are the inclination of the pendulum and the rate of falling.

The controller generates one output which is the horizontal force exerted on the cart.

6. Membership Functions

The input variables are then split into five linguistic areas namely Negative Large, Negative Small, Zero, Positive Small and Positive Large. The structure enables one to interpret continuous sensor data with a qualitative reasoning.

7. Rule Base Design

Twenty five fuzzy rules were executed. It is the rules that are developed out of human intuition and physical knowledge of the inverted pendulum system.

The fundamental control principle is that the cart should move in the same direction as the falling pole to reposition itself beneath the center of mass.

8. Controller Tuning

The controller was tuned by adding input scaling coefficients. The experimentally adjusted values of these coefficients were used to make a compromise between responsiveness and stability.

9. Simulation Results

The simulation results show that the pendulum can be stabilized for initial angles of approximately $\pm 30^\circ$. Within this range, the controller maintains the pole near the upright position.

For larger angles, the nonlinear dynamics become dominant and the controller may fail to recover, which reflects realistic physical limitations.

10. Control Surface Generation

A control surface was generated by evaluating the fuzzy controller across a range of angle and angular velocity values. The resulting data provides insight into the behavior of the fuzzy inference mechanism.

11. Discussion

The inability to stabilize all angles between -60° and $+60^\circ$ does not represent an error. It reflects the limited region of attraction inherent to inverted pendulum systems.

12. Conclusion

This project successfully demonstrates the application of fuzzy logic to a nonlinear control problem. The controller provides stable performance within a practical operating range and highlights both the strengths and limitations of fuzzy control.